Multi-omics survival biomarker for lung adenocarcinoma

Integrating more than 50,000 molecular features from fewer than 500 TCGA-LUAD patients: leakage-controlled benchmark, frozen signature and external validation

Overall survival (primary) in TCGA-LUAD, validated without refitting in GSE31210 and GSE68465. Prepared 6 November 2026.

Key findings

- **Internal criterion not met.** In repeated, leakage-controlled cross-validation, no way of integrating RNA, methylation and miRNA significantly improved Uno's C over age + sex + stage (0.636). The best, clinical + RNA, gave +0.012 [-0.026, +0.049].
- **A frozen clinical + 5-gene signature validates externally.** Harrell C was 0.715 in GSE31210 (226 tumours, 35 deaths) and 0.697 in GSE68465 (442 adenocarcinomas, 236 deaths); high-vs-low-risk hazard ratios were 4.36 and 2.90.
- **Most of the discrimination is clinical.** The clinical part alone reaches 0.68-0.69. The RNA part is indistinguishable from chance in GSE31210 (C 0.522) and small but real in GSE68465 (C 0.574; adds to the clinical part, likelihood-ratio p = 0.002).
- **Fully reproducible.** One command rebuilds every table and figure from the raw downloads; the regenerated signature is bit-identical to the frozen file.

1. Question and success criteria

The aim was a validated prognostic signature for censored overall survival from a very wide, noisy molecular matrix (RNA-seq, 450K methylation, miRNA) on a small cohort, with honest comparison to clinical baselines. Success criteria were fixed before modelling:

Criterion (fixed before the analysis)	Outcome	Evidence
Internal: multi-omics + clinical beats clinical-only on Uno C, CI excluding 0	**Not met**	Best: clinical + RNA, +0.012 [-0.026, +0.049]; shared-gene version +0.014 [-0.023, +0.051]
External: Harrell C > 0.60, lower CI bound > 0.50	**Met** (both cohorts)	GSE31210 0.715 [0.627, 0.792]; GSE68465 0.697 [0.661, 0.734]
External: high-vs-low risk HR > 1 with p < 0.05 (frozen cutoff)	**Met** (both cohorts)	GSE31210 HR 4.36 [2.21, 8.60]; GSE68465 HR 2.90 [2.23, 3.77]
External: calibration slope near 1 (rule set in advance: within 1 +/- 0.3)	**Met** in GSE68465, **not met** in GSE31210	GSE68465 1.24 [1.00, 1.47]; GSE31210 1.52 [0.81, 2.24] (CI includes 1)
Reproducibility: one command rebuilds tables and figures from raw data	**Met**	make reproduce; regenerated signature file bit-identical to the frozen one

2. Data and cohorts

Discovery cohort. TCGA-LUAD with the curated pan-cancer survival table (Liu et al. 2018). The table lists 641 LUAD rows; after keeping primary tumours (code 01), one sample per patient, and dropping missing or non-positive survival times, 506 patients remained. Requiring all three layers gave a **core cohort of 442 patients with 159 deaths** (RNA 503, methylation 447, miRNA 500 patients before intersection).

Scales. RNA was back-transformed from log2(count+1) and converted to log2 CPM; miRNA is log2 RPM; methylation beta values became M-values. Methylation probes on sex chromosomes were removed with a fixed annotation filter (385,809 autosomal probes kept), which removed a sex split that had dominated the second principal component (R-squared 0.69 to 0.05). Tissue source site was monitored as a batch variable with outcome-blind PCA.

External cohorts (named before any modelling). GSE31210 (Affymetrix GPL570): 226 tumours, 35 deaths, stage I-II, median follow-up 4.8 years; a sensitivity set of 204 excludes 22 flagged for incomplete resection or adjuvant therapy. GSE68465 (GPL96): 442 adenocarcinomas with follow-up (one more lacks a time), 236 deaths, median follow-up 3.9 years; stage was derived from pT/pN codes.

3. Methods

No leakage. The saved matrices are unit-transformed but unfiltered. Every data-dependent step ran inside the training fold: prevalence and missingness filters, imputation, univariate Cox score screening (top 1,000 genes, 300 miRNAs, 2,000 methylation probes), standardisation, penalty tuning, and the alpha path itself. Tuning was nested: elastic-net l1 ratio in {0.1, 0.5, 0.9} and 30 alphas, chosen by inner-CV Harrell C with the whole preparation re-fitted in every inner fold.

Evaluation. Repeated stratified 5x5 cross-validation, with identical seeded folds for every model so differences are paired. Metrics were Harrell's C and Uno's IPCW C truncated at 5 years. Confidence intervals and p-values use the Nadeau-Bengio correction for overlapping training sets. Clinical covariates (age, sex, stage) were unpenalised; omics were penalised.

Integration strategies compared: single layers; early integration (clinical + penalised omics); late integration (stacking inner-CV out-of-fold layer scores); intermediate integration (MOFA+ factors fitted in-fold, test patients projected, then Cox); and untuned random survival forests and gradient boosting as non-linear comparators.

Freezing and scoring. The signature model was refit on all 442 patients with the same tuning procedure that was cross-validated, restricted to the 11,753 genes measurable on TCGA and on both arrays (19% of RNA features, 56% of RNA variance; the restriction cost nothing in cross-validation). Coefficients, preprocessing and cutoff were written to a file whose hash was recorded before any external data were scored. Arrays were log2(x+1) transformed, the highest-mean probe was taken per gene, and each gene was z-scored within its own cohort; nothing was refitted, tuned or thresholded on external data.

4. Internal results

Table 1 compares every strategy on the same folds. The baseline is age + sex + stage. Omics-only models are below the clinical baseline, and no integration strategy beats it with a confidence interval excluding zero.

Model	Uno C (5y) [95% CI]	Uno C difference vs clinical	Harrell C [95% CI]	Harrell C difference vs clinical
Clinical baseline: age + sex + stage	**0.636 [0.586, 0.686]**	**+0.000 [+0.000, +0.000]**	**0.657 [0.607, 0.707]**	**+0.000 [+0.000, +0.000]**
Clinical: age + sex	0.469 [0.410, 0.527]	-0.168 [-0.235, -0.100]	0.471 [0.415, 0.527]	-0.186 [-0.251, -0.121]
Clinical: age + sex + stage + smoking	0.622 [0.572, 0.671]	-0.015 [-0.040, +0.011]	0.645 [0.600, 0.690]	-0.012 [-0.033, +0.009]
Clinical: stage only	0.625 [0.580, 0.670]	-0.011 [-0.043, +0.020]	0.659 [0.616, 0.702]	+0.002 [-0.021, +0.025]
RNA only	0.572 [0.511, 0.634]	-0.064 [-0.144, +0.017]	0.623 [0.569, 0.677]	-0.034 [-0.108, +0.040]
miRNA only	0.603 [0.537, 0.669]	-0.033 [-0.106, +0.040]	0.624 [0.558, 0.690]	-0.033 [-0.114, +0.047]
Methylation only	0.565 [0.510, 0.620]	-0.071 [-0.147, +0.006]	0.600 [0.539, 0.661]	-0.057 [-0.145, +0.030]
Early: clinical + RNA	0.648 [0.590, 0.706]	+0.012 [-0.026, +0.049]	0.686 [0.639, 0.733]	+0.029 [+0.005, +0.053]
Early: clinical + miRNA	0.626 [0.565, 0.686]	-0.011 [-0.045, +0.024]	0.660 [0.611, 0.708]	+0.002 [-0.033, +0.038]
Early: clinical + methylation	0.632 [0.577, 0.687]	-0.004 [-0.038, +0.031]	0.668 [0.622, 0.713]	+0.010 [-0.018, +0.039]
Early: clinical + all three layers	0.636 [0.584, 0.689]	-0.000 [-0.030, +0.030]	0.676 [0.630, 0.721]	+0.018 [-0.008, +0.044]
Early: clinical + RNA (shared-gene pool) = signature model	0.650 [0.596, 0.704]	+0.014 [-0.023, +0.051]	0.690 [0.648, 0.732]	+0.033 [+0.006, +0.060]
Late integration (stacking)	0.634 [0.581, 0.687]	-0.002 [-0.039, +0.035]	0.672 [0.625, 0.719]	+0.015 [-0.019, +0.049]
Gradient-boosted survival, clinical + omics	0.574 [0.523, 0.625]	-0.062 [-0.139, +0.015]	0.618 [0.565, 0.671]	-0.040 [-0.119, +0.040]
Random survival forest, clinical	0.607 [0.550, 0.664]	-0.029 [-0.070, +0.012]	0.645 [0.596, 0.693]	-0.013 [-0.054, +0.029]
Random survival forest, clinical + omics	0.579 [0.526, 0.632]	-0.057 [-0.135, +0.021]	0.625 [0.577, 0.673]	-0.032 [-0.106, +0.041]
MOFA+ factors only	0.568 [0.513, 0.624]	-0.068 [-0.149, +0.013]	0.605 [0.547, 0.662]	-0.053 [-0.135, +0.029]
Clinical + MOFA+ factors	0.613 [0.561, 0.665]	-0.023 [-0.080, +0.033]	0.651 [0.609, 0.693]	-0.006 [-0.062, +0.050]

Table 1. Repeated 5x5 cross-validation, identical folds, Nadeau-Bengio 95% intervals. Highlighted: the model frozen as the signature.

Reading. Harrell's C favours clinical + RNA (+0.029 to +0.033 with intervals just above zero), but Harrell's C was not the pre-specified metric and the gain does not survive correction for four combinations. Non-linear models did worse than the elastic net, and adding omics lowered the forest's Uno C from 0.607 to 0.579, consistent with many noisy features competing for about 159 events. Their hyperparameters were fixed in advance and not tuned.

Out-of-fold diagnostics (clinical + RNA vs clinical-only). Time-dependent AUC differences were +0.044 [+0.019, +0.074] at 1 year, +0.026 [-0.005, +0.054] at 3 years and +0.014 [-0.018, +0.043] at 5 years. These bootstrap intervals resample patients' out-of-fold predictions and ignore retraining variability, so they are narrower than the corrected intervals above and are descriptive only. Calibration slope was 0.97 [0.70, 1.24] for clinical + RNA and 0.84 for clinical-only; the integrated Brier score was 0.193 against 0.198 and 0.213 for a no-covariate model.

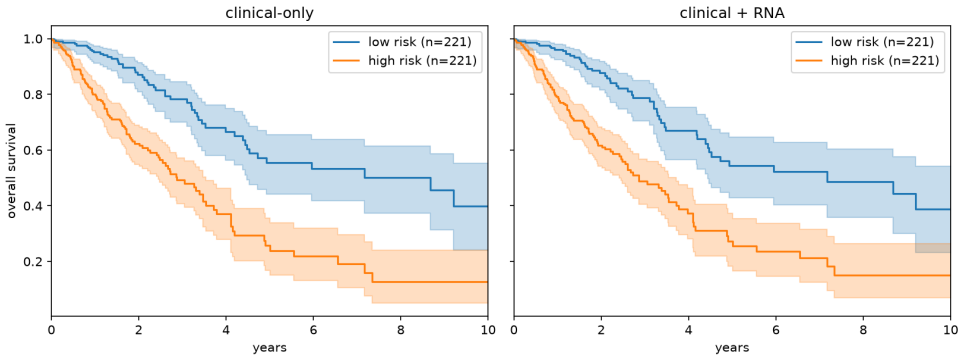


Figure 1. Out-of-fold median split of the risk score in TCGA (exploratory; the cutoff that counts is the frozen one). Hazard ratio high vs low 2.84 for clinical-only and 2.76 for clinical + RNA: the RNA adds no visible separation here.

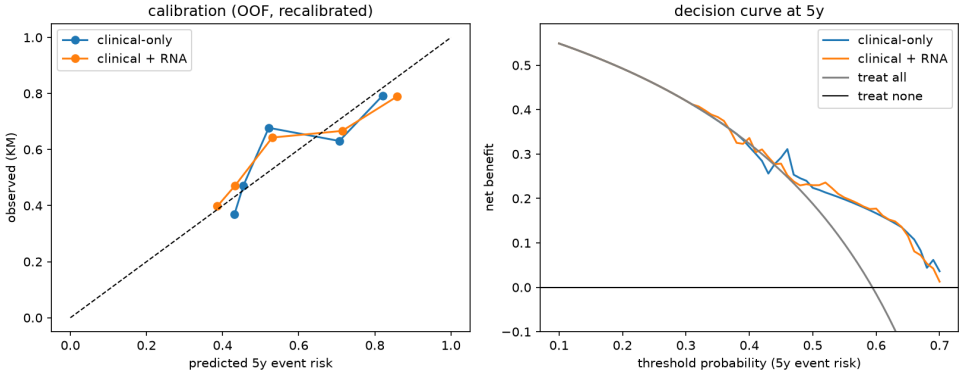


Figure 2. Left: calibration at 5 years (predictions recalibrated on out-of-fold scores, so absolute levels are optimistic; the comparison between models is fair). Right: decision curve at 5 years. Net benefit at a 40% threshold was 0.336 for clinical + RNA, 0.316 for clinical-only and 0.323 for treating everyone.

5. The frozen signature

Final fit on 442 patients and 159 events: l1 ratio 0.9, alpha 0.1049, inner-CV Harrell C 0.700 (the apparent training C of 0.725 is optimistic; the honest internal estimates are Harrell 0.690 and Uno 0.650). The frozen cutoff is the TCGA median linear predictor, -0.166.

Component	Term	Coefficient (log hazard per SD)
Clinical (unpenalised)	Age	+0.064
Clinical (unpenalised)	Male sex	-0.011
Clinical (unpenalised)	Stage II	+0.344
Clinical (unpenalised)	Stage III	+0.430
Clinical (unpenalised)	Stage IV	+0.291
RNA (elastic net)	DKK1	+0.148
RNA (elastic net)	TLE1	+0.116
RNA (elastic net)	SEC61G	+0.019
RNA (elastic net)	LDHA	+0.010
RNA (elastic net)	USP4	-0.009

Table 2. signature_v1: five RNA genes selected from 1,000 screened candidates. DKK1 and TLE1 carry nearly all the RNA weight (both were selected in 100% of cross-validation folds); both are known regulators of Wnt signalling.

6. External validation

The frozen signature was applied once to each external cohort. The full score, its RNA component and its clinical component were evaluated separately so the contribution of the transcriptome is visible.

Analysis	Harrell C [95% CI]	Uno C (5y)	HR per SD or high vs low [95% CI]	p
GSE31210 primary				
full signature	0.715 [0.627, 0.792]	0.682	1.88 [1.40, 2.53]	3.2e-05
RNA component only	0.522 [0.411, 0.631]	0.523	1.20 [0.86, 1.67]	0.275
clinical component only	0.693 [0.591, 0.786]	0.654	1.91 [1.43, 2.55]	1.4e-05
high vs low (frozen TCGA cutoff)			HR 4.36 [2.21, 8.60]	2.1e-05
GSE31210 sensitivity (204, flagged excluded)				
full signature	0.715 [0.608, 0.811]	0.681	1.89 [1.40, 2.57]	4.1e-05
RNA component only	0.556 [0.450, 0.671]	0.554	1.34 [0.94, 1.91]	0.110
clinical component only	0.681 [0.562, 0.787]	0.641	1.83 [1.36, 2.44]	4.9e-05
high vs low (frozen TCGA cutoff)			HR 4.40 [2.13, 9.07]	6.1e-05
GSE68465 primary (adenocarcinoma with follow-up)				
full signature	0.697 [0.661, 0.734]	0.698	1.95 [1.71, 2.21]	5.1e-24
RNA component only	0.574 [0.534, 0.611]	0.573	1.24 [1.09, 1.40]	0.001
clinical component only	0.691 [0.657, 0.725]	0.690	1.85 [1.63, 2.09]	2.7e-22
high vs low (frozen TCGA cutoff)			HR 2.90 [2.23, 3.77]	1.5e-15

Table 3. Frozen signature in external cohorts. HR per SD is for the continuous score; high-vs-low rows use the frozen TCGA cutoff (70/156 and 188/254 patients). With a within-cohort median split instead, the hazard ratios were 4.06 [1.84, 8.94] (GSE31210, 226), 3.81 [1.63, 8.88] (204) and 2.46 [1.88, 3.21] (GSE68465).

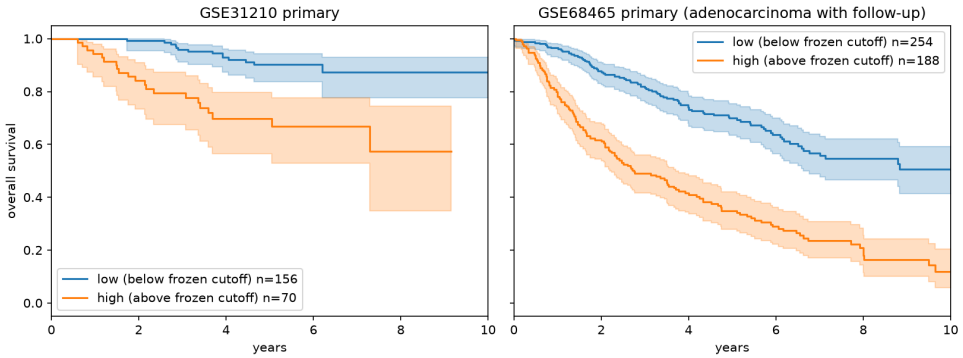


Figure 3. Overall survival by risk group at the frozen TCGA cutoff, GSE31210 (left) and GSE68465 (right).

What the external data show. The full score discriminates well in both cohorts (C 0.70-0.72), and the high-risk group has a clearly worse prognosis. Calibration slopes were above 1 (1.52 and 1.24; 1.62 in the 204-patient sensitivity set), which means predictions are too compressed rather than over-fitted. The clinical part alone gives 0.68-0.69, so the transcriptome adds 0.01-0.03. In GSE68465 the RNA part is significant on its own (HR per SD 1.24, p = 0.001) and adds to the clinical part (likelihood-ratio p = 0.002); the hazard ratio per SD was 1.92 [1.68, 2.19] after adjusting for adjuvant chemotherapy. In GSE31210 (stage I-II only, 35 events) the RNA part is indistinguishable from chance and its added value is not significant (p = 0.29 to 0.62), but that cohort has little power to detect a small effect.

7. Interpretation and limitations

- **Supported:** a clinical + RNA prognostic model replicates in two independent array cohorts; a weak independent RNA effect is supported by one large cohort and not detectable in the other.
- **Not supported:** that multi-omics integration improves on clinical-only prediction in TCGA at this sample size. The internal criterion was missed, and the confidence intervals (about +/-0.035) cannot exclude gains of up to about 0.05.
- GSE68465 stage was derived from pT/pN codes (no stage IV); three patients without stage were scored as stage I. GSE31210 contains only stage I-II patients, so the clinical block there separates little.
- The signature must be scorable on arrays, so it is built from genes shared by TCGA and the arrays. Methylation SNP and cross-reactive probes were not filtered; copy-number and somatic-mutation layers were not used.
- A mild proportional-hazards departure was seen for the clinical + RNA score (Schoenfeld p = 0.026); five-year restricted mean survival time differences between risk groups (about 410 days) were the same for both models.
- Incidents without effect on results: a first external run crashed on an index-alignment bug and was re-run after a mechanical fix; GSE31210 characteristic rows are ragged and are now parsed by label.

8. Reproducibility

make reproduce rebuilds every table and figure from the raw downloads in about one hour on a laptop: download, cohort, matrices and QC, clinical and single-omics baselines, integration, non-linear comparators and MOFA+, internal evaluation, signature regeneration, external validation and summary. The regenerated signature is written to a separate file and verified against the frozen one. On 6 November 2026 it was bit-identical (sha256 704878c704c0b13e3c1f4e3fb3e7671bf411eae7e13611f19342a47c8cd48041), and every results table matched the copy saved before the run. All seeds are fixed in the configuration file; the code, tests (18 passing) and a running log of decisions are in the project repository.